

Jonathan Vu

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EDUCATION

University of Maryland

Bachelor of Science in Computer Science + Math, 4.0 GPA

College Park, MD

August 2023 - May 2026

WORK EXPERIENCE

Software Development Intern

Remote

Geico

June 2025 - August 2025

- Built a full-stack developer management app with TypeScript/Javascript, React, and Django, streamlining workflows and reducing the barrier to onboarding new projects by over 50%
- Leveraged Large Language Models, Office 365, and the Slack API for automatic document generation and notifications
- Stored and queried thousands of data points with Microsoft GraphQL, PostgreSQL, and the Azure DevOps (ADO) API
- Used Pandas, Python, and SMTP servers to automate the sending and processing of ten-thousand email message processes
- Leveraged Insomnia, Figma, CI/CD, Power BI, and ADO to accelerate UI/UX design, planning, development, and testing

Data Architect Intern

Herndon, VA

Serco

May 2024 - August 2024

- Optimized source code for a Large Language Model, improving code structure and performance via efficient algorithms
- Utilized Python and Data Science to perform analysis on thousands of Task Orders and pinpoint growth opportunities
- Cleaned technical assets with Pandas and JSON structures, reducing Technical Debt by 33%
- Leveraged Power Automate, Logic Apps, and Azure to streamline processes for automated notifications and data processing
- Constructed Power BI Dashboards with internal data on contracts to analyze high-potential bidding opportunities

PROJECTS

EduAI | Bootstrap, PyTorch, Python

January 2025 - June 2025

- Developed a Bootstrap/Flask app integrating OpenAI API to dynamically create tests and study guides for personal learning
- Engineered a PyTorch Transformer model trained on 2 million data points to solve derivatives instantly
- Automated web content summarization with a BeautifulSoup scraper that parsed information from website URLs
- Streamlined my website with a SQLAlchemy ORM and OAuth backend to save user-generated notes, quizzes, and responses

VisionSynth | Python, OpenCV, Numpy

September 2024 - December 2024

- Worked with OpenCV to stitch images together in a Panorama with ANMS, RANSAC, and Feature Matching
- Trained a Gaussian Mixture Model for object detection and classification
- Implemented two view Stereo Reconstruction to rebuild 3D scenes from 2D images with Linear Triangulation
- Predicted the depth of objects with rectification and disparity, forming a depth map for objects

Crime Aid | First Place in Data Science at HooHacks 2024 | Sklearn, Pandas, TensorFlow

March 2024

- Formed an Arc-GIS full-stack application that displayed predicted criminal activity in a heat-map on the United States
- Leveraged a ReLU convolutional neural network in TensorFlow with 3 hidden layers, trained on 20k points
- Preprocessed data with Pandas and extracted features used to train the neural network

Rockfall | C#, Unity

June 2023 - August 2023

- Designed a 2D platforming game by creating maps and levels in C# with a team of 4 people
- Enhanced the player experience by creating 2 storyboard cut scenes and over 10 character scripts
- Improved the structure of the game by coding prefabs and assets used in designing sprites and levels

Quick Find | Python, Flask, Surprise

August 2022 - June 2023

- Developed a hybrid (content and collaborative) recommendation system utilizing Singular Value Decomposition (SVD), TF-IDF vectorization, and cosine similarity for accurate recommendations
- Programmed an interactive website in Flask that makes recommendations of places for users to visit in 5-10 seconds with scikit-learn and Surprise, based on data points sourced from Yelp
- Leverage the Google Places API and Geopy to search for nearby places, allowing users to find specific locations
- Documented and presented my results and research through a twenty page research paper and presentation

TECHNICAL SKILLS

Languages: Python, Dart, Java, C#, C++, C, JavaScript/Typescript, SQL, HTML/CSS, MIPS Assembly, Rust, OCaml

Frameworks: Node.js, Flask, Django, Bootstrap, Flutter, React

Developer Tools: Azure, Android Studio, Power Automate, Logic Apps, Git, Power BI, Google Cloud Platform, Linux

Libraries: Pandas, NumPy, Matplotlib, Scipy, OpenCV, Surprise, PyTorch

ACHIEVEMENTS

National Merit Scholar: Received the national merit scholarship from the University of Maryland in 2023

UMD Presidents Scholarship: Received a 40k scholarship at the University of Maryland in 2023